

Chapter 12, Problem 75P

Problem

A man has a body resistance of $600\ \Omega$. How much current flows through his ungrounded body:

(a) when he touches the terminals of a 12-V autobattery?

(b) when he sticks his finger into a 120-V light socket?

Step-by-step solution

Step 1 of 2

$$(A) \quad I = \frac{V}{R} = \frac{12}{600} = 20\text{ mA} .$$

Step 2 of 2

$$(B) \quad I = \frac{V}{R} = \frac{120}{600} = 200\text{ mA} .$$